Lecture 3 — Statistical Testing, continued

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IMG

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Part I — Associations between categories

The contingency table

Are two categorical variables associated?

- Cross-classify the counts into a contingency table with row and column marginal sums (the Lady tasting tea: guessed vs. actual).
- Shared null hypothesis: the two variables are not associated (independent).
- Two routes: an exact combinatorial test (Fisher) or a large-sample approximation (chi-squared).

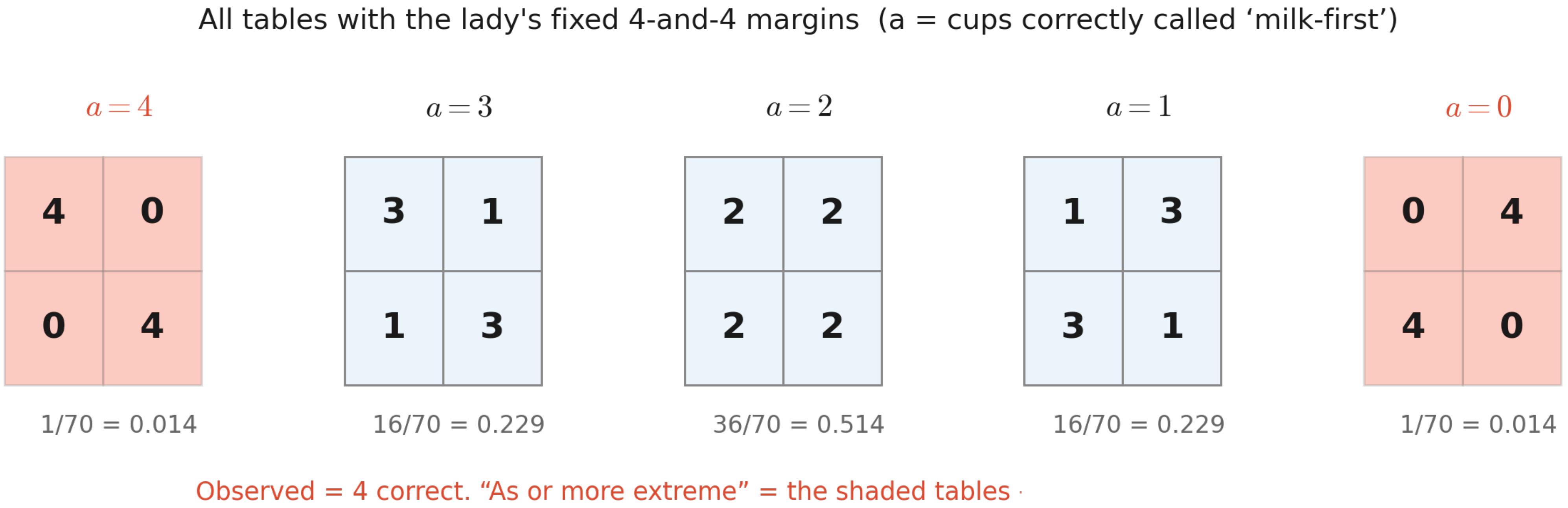
	detected as milk before	detected as milk after	marginal sums
milk before	3	1	**4**
milk after	1	3	**4**
marginal sums	**4**	**4**	**8**

Counts in a table — is the pattern more than chance?

Fisher's exact test

Count the tables, exactly.

- p-value = the fraction of all tables with the same margins that are as or more extreme than observed — pure combinatorics, no approximation.
- Reports an odds ratio summarising the strength of association.



Exact for any sample size — but it doesn't scale.

Chi-squared test of independence

Compare observed counts to what independence predicts.

Observed counts (E = expected under independence)

	Worse	Same	Better	Σ
Treat A	**12** E=18	**31** E=33	**57** E=49	**100**
Treat B	**24** E=18	**35** E=33	**41** E=49	**100**
Σ	**36**	**66**	**98**	**200**

Expected count from the margins

$$E_{ij} = \frac{(\text{row sum}_i)(\text{col sum}_j)}{N}$$

Statistic

$$T = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} \sim \chi^2 \text{ with } df = (\text{rows} - 1)(\text{cols} - 1)$$

Fisher's large-sample cousin — an approximation that scales.

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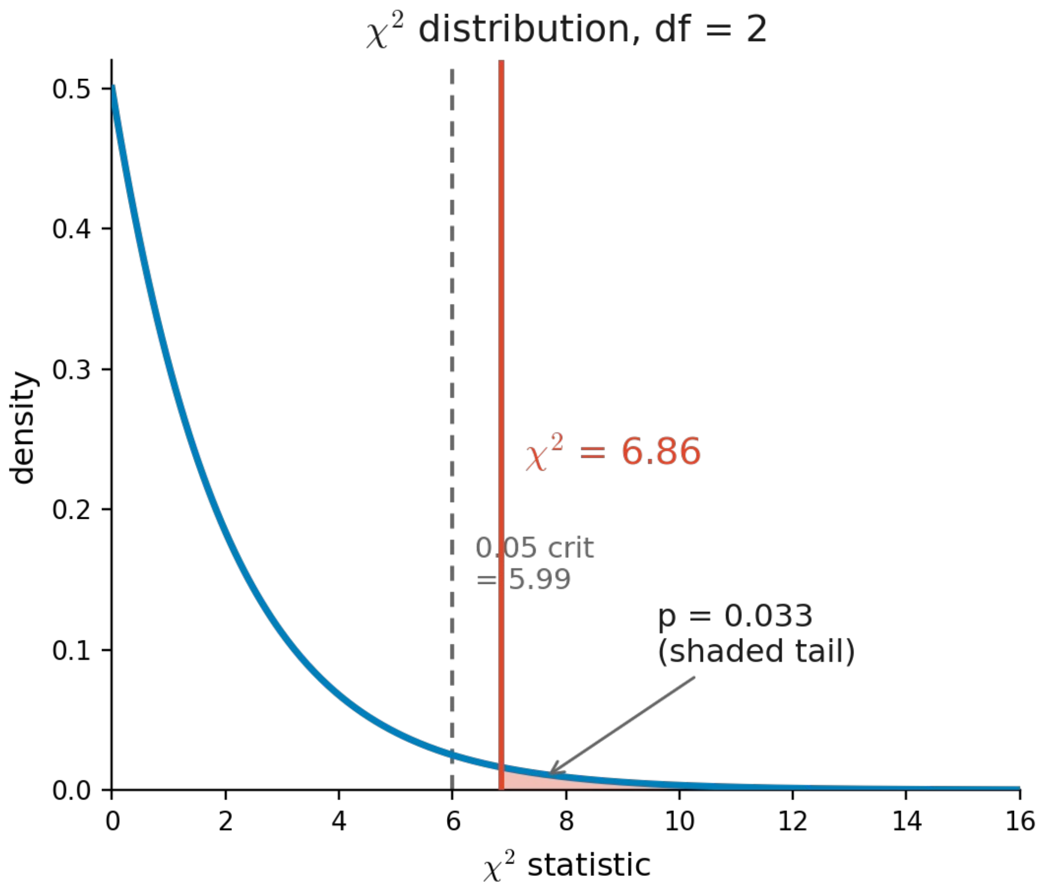


Fisher's large-sample cousin — an approximation that scales.

Part II — Comparing whole distributions

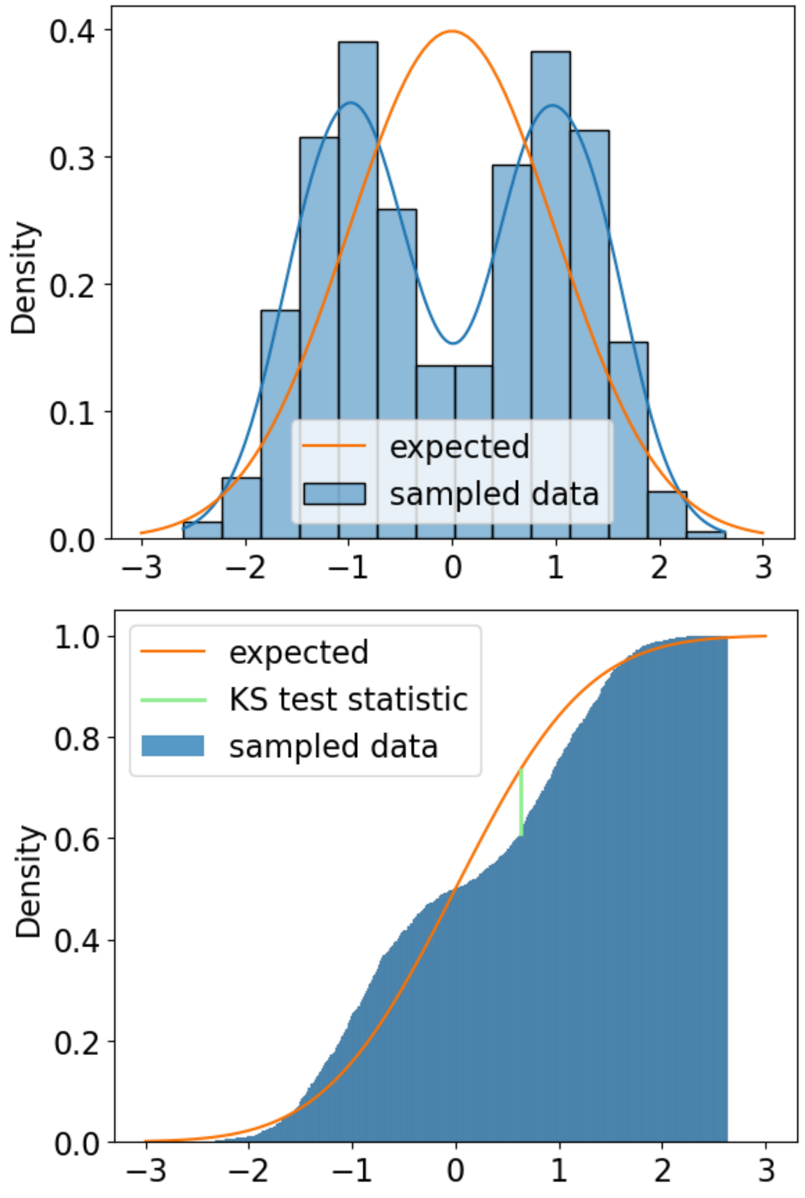
The Kolmogorov–Smirnov test

Do two samples come from the same distribution?

- Nonparametric; compares entire CDFs
- a sample against a reference (1-sample)
- or two samples against each other (2-sample).

- Statistic = the largest vertical gap between the two cumulative curves.

- Unlike the t-test (location only), it detects differences in location, scale, or shape.



A generalist: sees any difference, but weakly.

Part III — Comparing more than two groups

First, check the variances: Bartlett (and Levene)

The tools ahead assume equal variances — test that first.

- Bartlett's test tries to reject the null hypothesis of homoscedasticity = equal variances across the m groups

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_m^2$$

- statistic ~ χ^2 with m−1 df
- Bartlett assumes normal data; if normality is doubtful, use Levene's test (more robust).

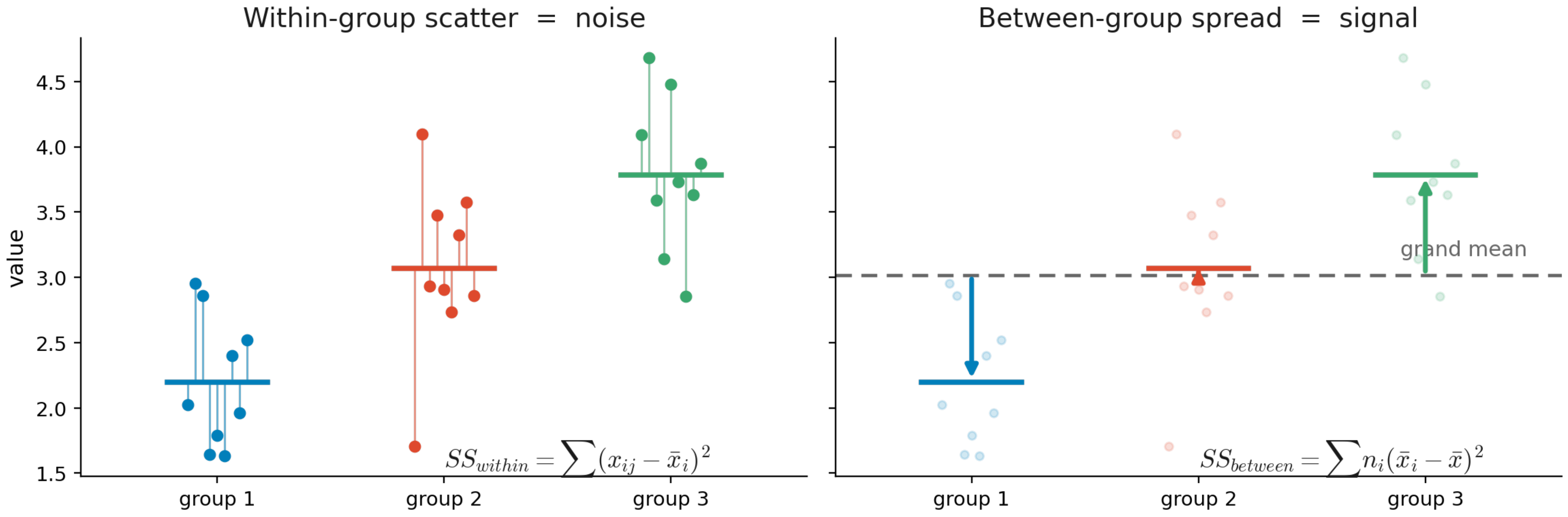
- C.f. heteroscedasticity

Equal variances? The answer picks which ANOVA to run.

One-way ANOVA

The t-test, generalised to many groups.

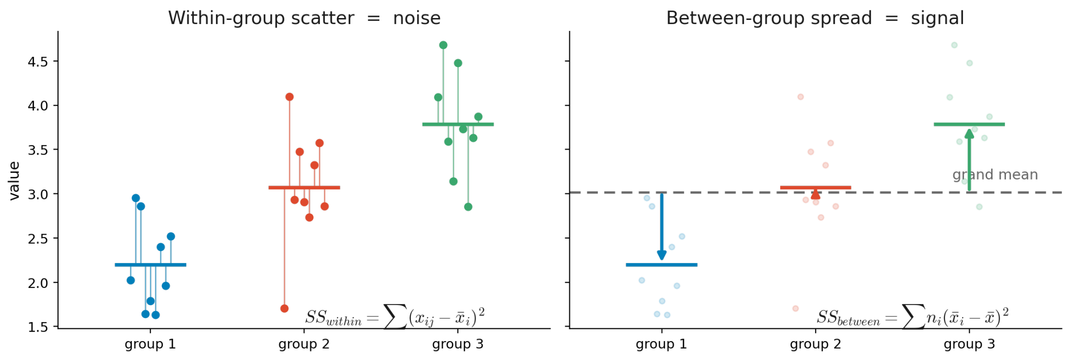
- Compares the means of a numerical variable across groups set by one categorical variable.



Signal / noise: between-group spread over within-group scatter.

One-way ANOVA

- Compares the means of a numerical variable across groups set by one categorical variable.
- H_0: m group means are equal



- Splits the total variation into between-group and within-group parts:
- Total variation: every point's deviation from the grand mean

$$SS_{\text{total}} = \sum (x_{ij} - \bar{x})^2$$

$$SS_{\text{total}} = SS_{\text{between}} + SS_{\text{within}}$$

- Between group: each group mean vs. the grand mean (signal)
- Within group: each point vs. its own group mean (noise)

$$SS_{\text{between}} = \sum_i n_i (\bar{x}_i - \bar{x})^2$$

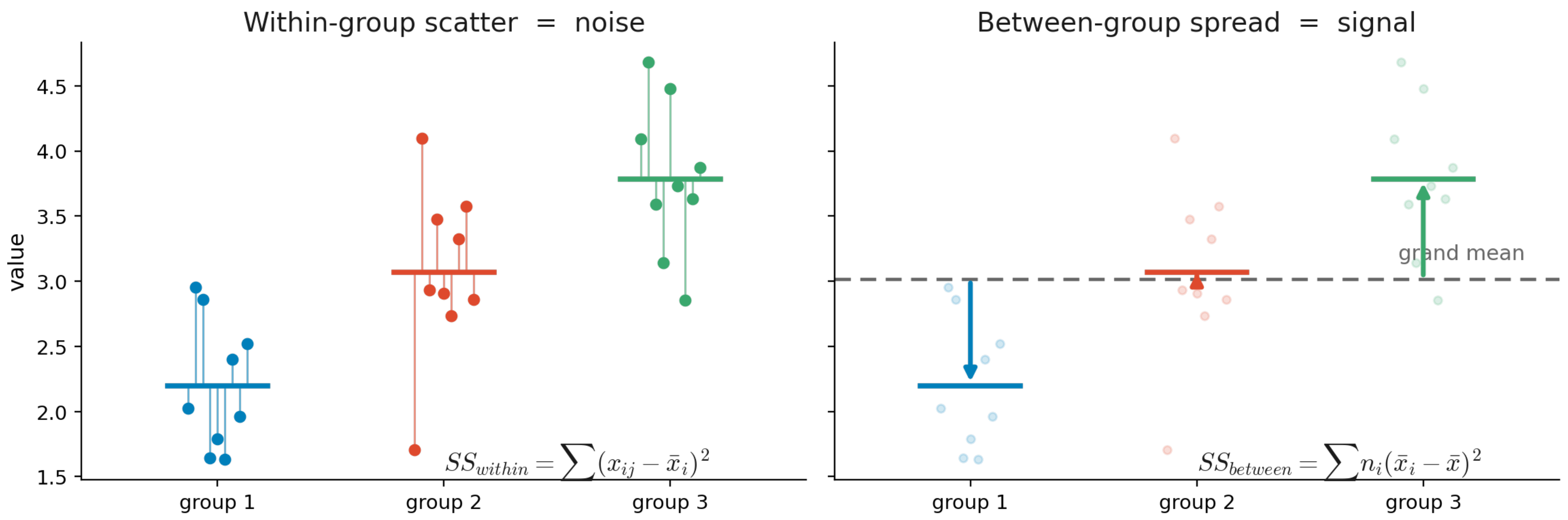
$$SS_{\text{within}} = \sum_{i,j} (x_{ij} - \bar{x}_i)^2$$

Signal / noise: between-group spread over within-group scatter.

One-way ANOVA

- Compares the means of a numerical variable across groups set by one categorical variable.
- H_0: m group means are equal
- The statistic is the signal over noise ratio $F = \frac{SS_{\text{between}}/(m-1)}{SS_{\text{within}}/(n-m)}$

- Large F $\Longrightarrow$ groups differ
- Follows F-distribution.



Signal / noise: between-group spread over within-group scatter.

ANOVA assumptions & alternatives

Three assumptions — and a backup for each failure.

- Assumes normal subpopulations, equal variances, and independent observations.
- Normal but unequal variances → Welch's ANOVA or Alexander–Govern test.
- Non-normal → Kruskal–Wallis H test, a rank-based test of median equality.

Match the test to the assumptions that actually hold.

Which groups differ? Tukey's HSD

ANOVA says "someone differs" — Tukey says who, without inflating error.

- Run after a significant ANOVA: all pairwise differences at once, holding the false positives at α using family-wise error rate.
- Yardstick is MS within — ANOVA's pooled estimate of the common noise σ^2 over all n−m residual df, so it is stable and on the same scale.

$$q = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{MS_{\text{within}}/n}}, \qquad MS_{\text{within}} = \frac{SS_{\text{within}}}{n-m}$$

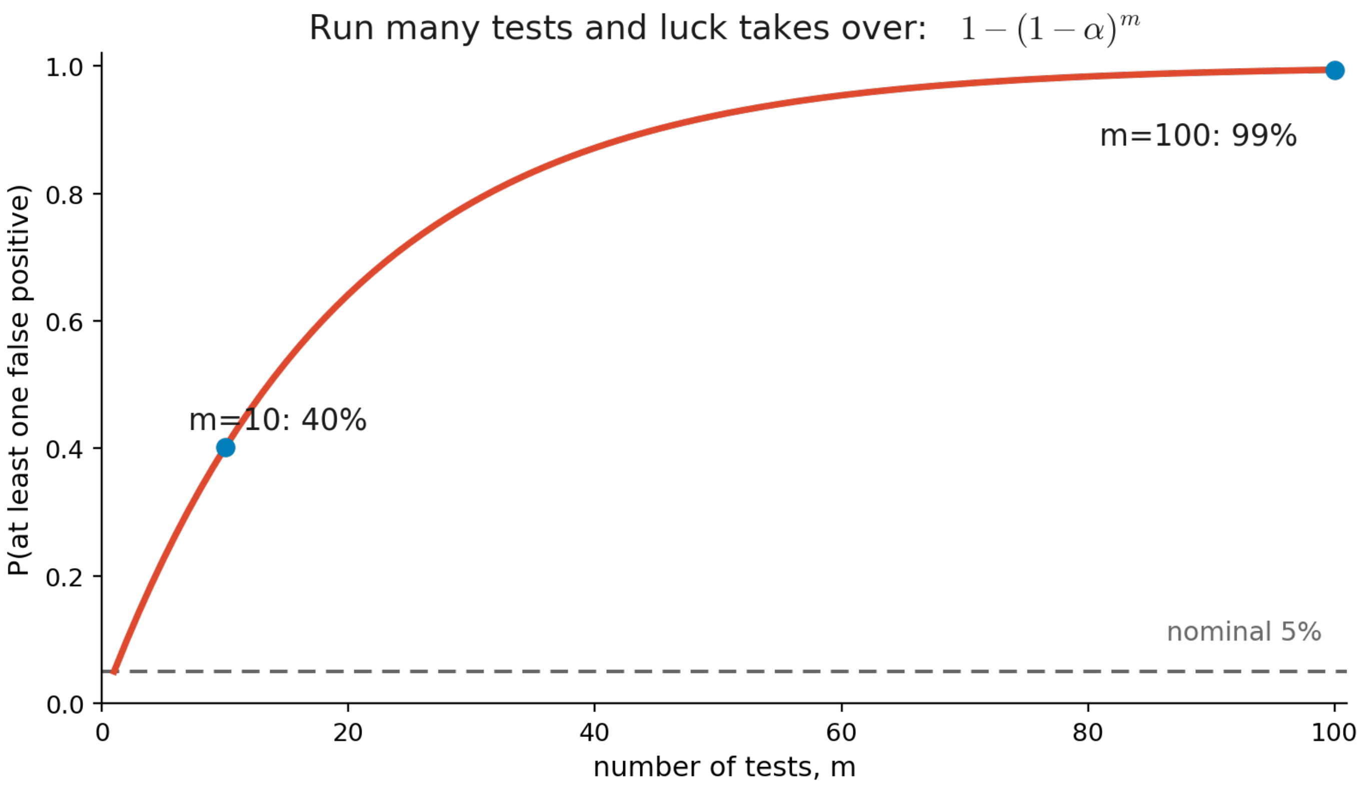
- q is compared to Studentized Range distribution $q_{\alpha,\, m,\, n-m}$ — the distribution of the largest difference among m means (, which is what makes the FWER correction automatic).
- Honestly Significant Difference (HSD) = the smallest mean gap that clears that critical value; pairs farther apart than the HSD are significant.

It reuses ANOVA's MS_within as the common yardstick (same equal-variance assumption)

Part IV — Running many tests at once

The multiple-testing problem

Run enough tests and you'll be “significant” by luck.



5% per test ≠ 5% overall.

Controlling error: FWER vs FDR

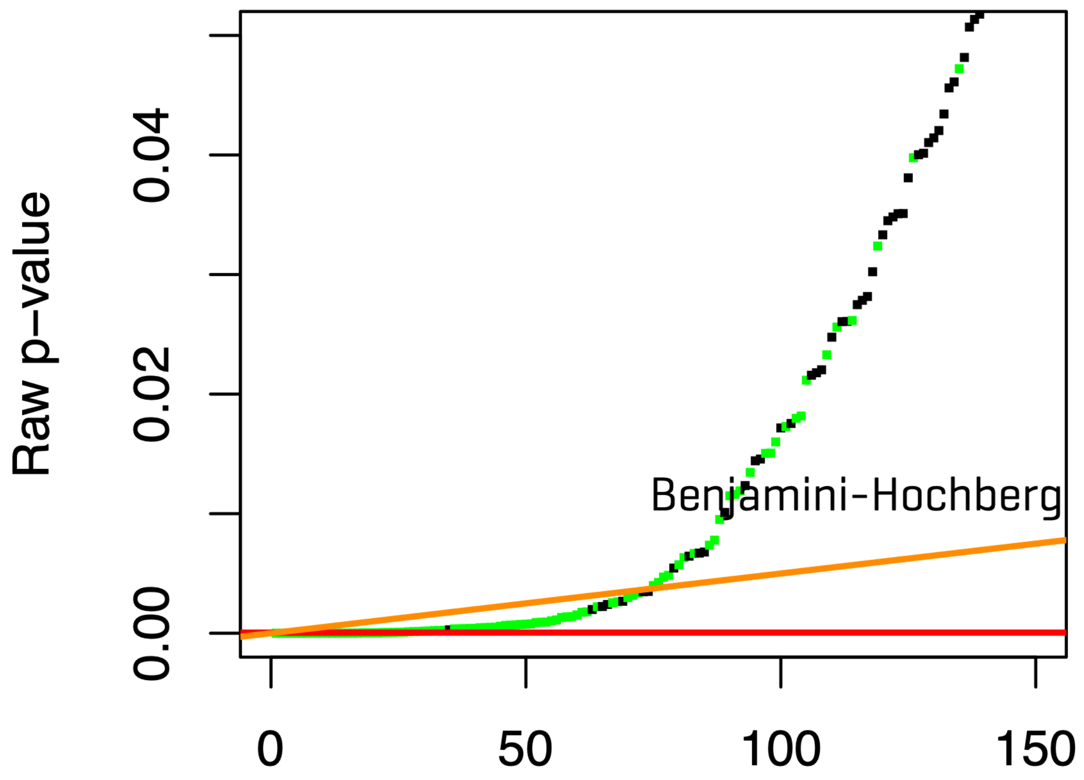
Two different things you might choose to control.

FWER (family-wise error rate)

- Controls Prob(any false positive).
- Bonferroni–Šidák: call significant only below $p < \alpha/N$
- FWER is strict (few false positives, many misses)

FDR (false discovery rate)

- Expected proportion of false positives among hits.
- Benjamini–Hochberg.
- FDR keeps power for large screens



Bonferroni for certainty; Benjamini–Hochberg for discovery.